Selected Experience: A Preregistered Test of Reputation as a Data Filter Under Recursive Training

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Abstract

Recursive training on synthetic data erodes the tails of the original distribution. Verifier-based selection is an established remedy, and weighting assessors by their reliability is an old technique. What remains open is whether domain-scoped reliability weights, earned once against a key, improve a validity filter under recursion. This paper preregisters, freezes, and runs a test of that question. On a two-dimensional Gaussian mixture with simulated validators, it asks whether a reputation-weighted validator pool, used as a data filter, achieves higher capped per-component rare-mode retention than full replacement, the published ten percent real-data control, and a likelihood filter, at an equal candidate budget, and, separately, whether any difference is attributable to the weights rather than to pooling. The test is distinct from the parent paper on Reinforcement Learning from Reputation Feedback (RLRF), whose proposed consensus-filtered training format is untested and whose own selection experiment weights assessors by consensus and selects once, and it is not the end-to-end policy-training comparison that paper calls for. A replication of the published language-model protocol served as the gate for reading any outcome. The published collapse result did not reproduce under the frozen rule. The replication failed. Substrate G is not read, its outcomes stay sealed, substrate F is blocked as well, and the paper reports no filter result. The fact-corpus substrate remains procedure-invalid. Its nine registered tests stand at $p = 1$ with the procedure-invalid label and no other.

Keywords: model collapse, recursive training, synthetic data, reputation systems, data filtering, preregistration, validator pools, Reinforcement Learning from Reputation Feedback

JEL Classification: C15, C18, C63, D83, O33.

I. From Mechanism to Measurement

Generative models now train on the record their predecessors wrote. When a model is fit to samples drawn from an earlier model, and that loop repeats, the tails of the original distribution disappear first and later generations converge toward a point estimate with very small variance (Shumailov et al. 2024). The low-probability regions of the data are the first casualties. Image generators that consume their own output without enough fresh real data progressively lose quality or diversity (Alemohammad et al. 2024), and the scaling laws that made large models predictable change form once synthetic data enters the corpus (Dohmatob et al. 2024). As artificial-intelligence-generated content proliferates, it dilutes the diversity of the text available for later training, potentially degrading performance across generations (Kaal 2025).

The literature offers two established lines of remedy. The first keeps real data in the loop. The second selects: synthetic data that passes a verifier can sustain performance where unfiltered recursion fails (Feng et al. 2025). Neither asks how the judgments behind a validity filter should be weighted when several assessors of uneven and domain-specific reliability supply them.

The paper treats that question as one of mechanism design. Its working hypothesis is that collapse on the replacement protocol is, at least in part, a selection failure rather than a generation failure. Whoever writes the filter writes what the next generation learns. Reinforcement Learning from Reputation Feedback (RLRF) proposes an alignment mechanism in which an economically incentivized validation pool, whose members stake non-transferable reputation, replaces the human annotator panel (Kaal 2026b). Reputation there rises when a vote matches the pool's reputation-weighted consensus and falls more steeply when it does not (Kaal 2026b), and it is scoped by domain, so that standing earned in one field confers no weight in another (Kaal 2026a). The RLRF paper does not test its mechanism against model collapse, and it poses no filter inside a recursive replacement loop as a remedy. That use of the mechanism is this paper's question.

The question is open on the evidence. The RLRF paper reports a cohort in which a party's record on checked work predicts its accuracy on unchecked work, but its best-record selector's margin over shipping whatever passes the checks in hand is 0.037, with an interval containing zero, and its own statement of limits calls RLRF an untested mechanism proposal (Kaal 2026b). Its separate selection experiment did not show that consensus-weighted selection outperformed equal-weighted selection, and its attribution test placed the observed difference at the 53.4th percentile of random reassignments of the weights (Kaal 2026b). A weighting rule earns no credit as a collapse remedy until it beats the simplest real-data remedy on a collapse protocol.

Three commitments discipline the answer. First, the paper names the limit of its own mechanism before any data exist. A validity filter judges whether an output is correct, not whether it is rare. Where rare content is as checkable as common content, such a filter can change rare composition only through the generator's own error profile. If collapse proceeds by under-sampling rare modes, no validity filter restores them, and a reputation-weighted filter is no exception. The experiment therefore includes an oracle validity filter as a premise. Second, the outcome is chosen to register the loss collapse inflicts. Total rare-region mass behaves close to a martingale under resampling: individual rare components drift and die while the aggregate stays near its starting value. The primary outcome therefore caps each rare component's retained mass at its original level and averages over components. A component that dies counts as a loss. A component that grows cannot offset it. Third, the design is built so that its author arranged not to read the outcome early, and so that specified later adjustments would be detectable.

The paper makes three contributions, each narrower than the mechanism it descends from. It poses, and preregisters a test of, whether domain-scoped reliability weights earned once on a disjoint key improve a validity filter under recursive replacement, against full replacement, real-data retention, and a likelihood filter, and against equal-weight pooling and a single verifier. It contributes a design for testing validity filters

under recursion: a capped per-component outcome, an oracle premise that gates attribution, a bounded-exposure real-data comparator, and a coin-flip validator family whose leak or acceptance failure removes confirmatory status. And it implements the preregistration and blind-analysis standards strictly in its blinding and analysis controls, with two departures from its frozen procedure that Section IV discloses. That record is evidence about the conduct of the study, not about the filter.

Relation to the RLRF paper. This paper is a follow-on to the RLRF paper (Kaal 2026b). That paper proposes the mechanism, reports a measured cohort, a parameter sweep, and a separate selection experiment, and trains no policy. Relative to the parent paper, the contributions above come to three additions. The first is a preregistered and frozen test of one use of the mechanism, as a filter on the training data of a recursive replacement loop. The second is an implementation of that test. The published collapse result did not reproduce under the frozen rule. The replication failed. Substrate G is not read, its outcomes stay sealed, substrate F is blocked as well, and the paper reports no filter result. The third is a statement, in Section II, of how the design bears on three of the parent paper's open questions and what it leaves open. Section VII reports the failed replication. Section VIII gives the registered reading of that outcome. Sections IX and X state the resulting limits and conclusion. The paper does not test the parent mechanism as a whole, and it does not supply the end-to-end policy comparison the parent paper calls for.

II. What the Literature Already Establishes

Every component of the experiment has a literature older than the experiment. This section proceeds by concession: it names the closest work, states what that work establishes, and isolates what is left.

Which collapse. Shumailov et al. (2024) showed tails disappearing first across language models, variational autoencoders, and Gaussian mixtures. Theory located the boundaries: iterative retraining is stable when the initial model approximates the data distribution well enough and the share of clean real data in each round is large enough (Bertrand et al. 2024), purely synthetic training cannot avoid collapse while mixing in real data bounds the synthetic share that can (Seddik et al. 2024), and synthetic data cuts or narrows the tail of the data distribution and changes the form of scaling (Dohmatob et al. 2024). The regime matters most. Replacing real data with each generation's synthetic data collapses, while accumulating real and synthetic data keeps error bounded (Gerstgrasser et al. 2024; Kazdan et al. 2025), and some argue that the catastrophic reading misdescribes realistic conditions (Schaeffer et al. 2026). This paper runs fixed-size replacement, the protocol on which collapse was first shown, and has no accumulation arm, so no result here bears on whether a filter adds anything once real data accumulates. Its outcome defines collapse as finite-sample extinction of low-weight modes.

Real data as a remedy. Retaining ten percent of the original data slows the damage (Shumailov et al. 2024), and accumulating all real data keeps test error bounded in the settings studied (Gerstgrasser et al. 2024; Kazdan et al. 2025), although even a synthetic share as small as 1 per 1,000 can stop larger training sets from helping in regression (Dohmatob et al. 2025). A filter that claims to address collapse must therefore beat real-data retention, not only unfiltered replacement.

Selection as a remedy. The closest prior work is Feng et al. (2025), who show that verifier-selected synthetic data can prevent collapse, with Gaussian-mixture theory and experiments comparing oracle and weak verifiers with self-selection and random selection. The paper concedes the core of that result: verifier selection is an established remedy, and the oracle ceiling and the Gaussian setting are established. Later work maps the limits. An imperfect external verifier drives the model toward the verifier's own knowledge center (Yi et al. 2026). Synthetic data helps only when the loop is information-open, shaped by external signals that inject task information, and a coarse signal such as binary correctness is domain-general (Li, Sun, and Deng

2026). Self-verification by internal confidence can prevent collapse even in fully synthetic regimes (Fu et al. 2025). Curation acts as implicit preference optimization that amplifies the curator's biases (Ferbach et al. 2024), several heterogeneous rewards can instead preserve diversity (Falahati et al. 2026), and verifiers that see only fragmented slices of the reference can accelerate tail loss (Qiao et al. 2026). The pool tested here applies one domain-general validity bit, so what domain scoping can govern is whose judgment of that bit counts, not what the bit says.

Selection on rarity. Filters that select on surprise or entropy, not validity, work where they work because they see what a validity filter is forbidden to see: a high-surprise filter mitigates collapse about as well as human-data baselines (Gambetta et al. 2026), and an entropy-rate filter reports diversity gains where a log-probability filter gives no significant benefit (Mitchell 2026). Against that literature, a null here has a literature rather than a caveat.

What the weight already is. Weighting assessors by estimated reliability is old (Dawid and Skene 1979; Whitehill et al. 2009), and multiplicative weights supply its online form (Littlestone and Warmuth 1994; Freund and Schapire 1997). Peer prediction makes honest reporting an equilibrium without a key (Miller, Resnick, and Zeckhauser 2005; Prelec 2004). The RLRF paper concedes its weight is a member of this family (Kaal 2026b). In this experiment the concession is stronger: reputation is settled once against a key before training, so the pool is supervised reweighting from a disjoint key. The paper does not compare it with a key-free latent-variable aggregator.

Goodhart. Optimizing a proxy eventually degrades the target (Gao, Schulman, and Hilton 2023), and curated loops optimize their curator (Ferbach et al. 2024). This is the channel the RLRF program claims reputation can resist. This study does not open it: its validators are stationary, its stake has no external value, and its weights are fixed before generation 1.

Preregistration. Preregistration, registered reports, and blind analysis are established standards (Nosek et al. 2018; Chambers and Tzavella 2022; MacCoun and Perlmutter 2015). The paper claims no invention there.

What the parent paper already reports. Four passages of the RLRF paper bear on the question here, and the present study is distinct from each.

First, the RLRF paper proposes, without implementing or testing it, a training format in which candidates that received an UP resolution with high reputation-weighted consensus become training targets, and it lists the resulting flywheel, in which better pools yield better training data, as untested at every stage (Kaal 2026b). That format is the parent paper's nearest construct to a data filter. It weights by pool consensus rather than by a key, sits outside any collapse protocol, and has not been run. The filter here weights by a key, operates inside a fixed replacement protocol, is scored by a capped rare-component outcome, and has been run on its Gaussian substrate, with outcomes sealed.

Second, the RLRF paper's Experiment T compared consensus-weighted with equal-weighted selection by a panel of seven small open instruct models choosing among candidate responses on a two-constraint instruction-following task that the paper calls easy (Kaal 2026b). The observed difference was about 0.002, its sign reversed in the paper's own sensitivity check, its interval contained zero, attribution was negative, and the paper states that the experiment does not test the mechanism. Both experiments ask whether weighting assessors adds anything to pooling them, but they differ on every axis that defines a test. Experiment T weights by consensus, selects once, and uses model assessors whose reliability is not set by design. Here the weights come from a key on disjoint items, the filter governs the training data of fifty generations of recursive replacement, and the validators are simulated with a reliability structure set by design, so that in two of the three required families the weights could matter, as a feature of the design and

not an expectation about the outcome. Experiment T is not revisited, and a result in one experiment would not answer the question posed by the other.

Third, the RLRF paper leaves open whether reputation weights come to track validator accuracy rather than agreement with the pool, and whether weights earned on keyed tasks transfer to unkeyed ones (Kaal 2026b). The ladder of Section V asks a narrower form of both. Its third rung asks whether the weights track accuracy on held-out keyed items, but every update here is driven by the key, so the contrast with agreement with the pool never arises. Its fourth rung asks whether weights earned on keyed rows improve the validity of candidates that no key settles, but the simulated validators' accuracy depends only on the tag, so the keyed rows are informative about the candidates by construction, and the rung checks whether the pipeline carries that information into what the pool accepts, not whether real validators would show the same transfer. In the homogeneous family, where the weights have nothing to find, the rung is a bound on attribution and not a prediction. The parent paper's measured cohort concerns the records of real model parties rather than pool weights, and its parameter sweep uses simulated validators that the paper itself says cannot validate the mechanism (Kaal 2026b). The simulated families here carry the same limit, and their errors are independent by construction, so validator independence is not examined either.

Fourth, the RLRF paper names, as prior to the six open problems it then lists, an end-to-end comparison in which policies are trained under a reputation-weighted signal, under an annotator or model-rater baseline, and under a visible-check selector, are evaluated on held-out work, and have the reputation weighting ablated separately (Kaal 2026b). The present design is not that comparison. Its pool supplies no reward signal and trains no policy, and its equal-weight contrast ablates the weights inside a data filter rather than inside policy training, so it does not supply the ablation that paper asks for. It also runs no Full Vote, the staked and slashable second round of the RLRF protocol, and its single keyed update is not a stake settlement.

What is left is a comparative identification question none of this work answers: on the replacement protocol, under an outcome that counts the death of a rare component, with every arm denied rare-set membership, does a validity filter whose validators carry domain-scoped weights earned on a disjoint key retain more of the rare components than full replacement, real-data retention, and a likelihood filter, and does any advantage survive ablation to equal-weight pooling and to a single verifier?

III. The Mechanism Under Test

The two pools at the center of the design, the reputation-weighted pool and the equal-weight pool, apply every rule in common except one: the weight a validator's vote carries on a domain, and whether that weight was earned. Every rule below is the frozen design's rule for substrate G, the only substrate on which the filter is tested.

What any filter may know. Validators judge validity only. No validator, filter, or score is asked about rarity, novelty, frequency, diversity, or coverage. The outcome is computed by an evaluator alone, from a sealed reference holding rare-set membership and component identity, which no filter reads. The oracle alone receives one bit per candidate from it, the candidate's true validity verdict. A point x is valid when its squared Mahalanobis distance to the nearest component mean satisfies $d_M^2(x, \mu_k) \leq \chi^2_{2,\,0.95} \approx 5.991$, so validity is balanced between rare and common regions to within 0.02, as a check before any run confirmed. Validators also see a tag, the nearest of four hash-drawn centers, chosen so that $|P(\text{rare} \mid t) - P(\text{rare})| \leq 0.04$ for every tag t.

The pool. Five validators are seated per candidate from a roster of twenty. Each votes independently. A candidate c is accepted when its weighted valid share $s(c) = \sum_i w_i v_i(c)$ exceeds $\theta = 0.5$, where $v_i(c) \in \{0, 1\}$ is validator i's vote. A candidate with $|s(c) - \theta| \leq 0.1$ is contested and receives four more validators, and

the recomputed share binds. Each vote stakes one reputation unit, a constant with no external value. The RLRF paper specifies a two-round protocol with argumentation between rounds (Kaal 2026b). This study keeps the weighted vote and the stake and replaces the two rounds with a single assessment and a threshold-triggered second look. Every drawing arm samples its own 4,000 candidates per generation and trains on 2,000 rows. Validator compute is not equal across arms: the pool seats five validators per candidate, nine when contested, and the single verifier one. A threshold arm that accepts too few fills the shortfall uniformly from rejected candidates, and an arm whose minimum over generations of the seed-mean fill-free share, $\min(\text{accepted}, N)/N$ with $N = 2{,}000$ training rows per generation, falls below $a_{\min} = 0.98$ in a family has its tests there invalidated.

The weight. The update is equation (3) of the RLRF paper, an asymmetric multiplicative rule. On tag t, reputation updates as $r_{i,t} \leftarrow r_{i,t}(1+\alpha)$ when the vote matches the revealed key and $r_{i,t} \leftarrow r_{i,t}(1-\gamma)$ when it does not. The RLRF paper requires only $\gamma > \alpha$, and this design fixes $\alpha = 0.05$ and $\gamma = 0.10$. Each reputation starts at $r_{i,t}^{(0)} = 1$, and after each update the floor $r_{i,t} \geq 0.01$ is applied. Weights are $w_i = r_{i,t} / \sum_j r_{j,t}$ over the seated validators. Any weight above 0.5 is capped at 0.5, and the excess is redistributed among the other seated validators in proportion to their weights, so that $\sum_i w_i = 1$. The RLRF paper states the rule for unkeyed events and allows keyed settlement. This study uses only keyed settlement, once, after the generation-0 fit on real data and before generation 1. No candidate that enters training ever updates a reputation. The keyed record holds 192 real rows, stratified so that it teaches the pool who judges validity well on which tag and nothing about where the rare components lie beyond the bounded tag leakage. Those $K_G = 192$ rows are the pool's entire exposure to real data, below the $\lfloor 0.1N \rfloor = 200$ real rows the ten percent arm receives in one generation and far below the $N(1 - 0.9^{50}) \approx 1{,}990$ distinct real rows it trains on over the run.

The validators. Twenty simulated validators vote the true validity bit, flipped independently with probability one minus their accuracy on the candidate's tag. Family U is homogeneous at 0.75. In family H each validator is reliable at 0.90 on two hash-drawn tags and at 0.60 on the other two, so domain-scoped reputation has something to find. Family Mn has five validators at 0.90 and fifteen at 0.70, a competent minority. The three share a mean accuracy of 0.75. Family Z votes by coin flip. It is a negative control: a coin-flip pool knows nothing about validity, and if it meets the registered leak criterion or fails its acceptance rule, both substrates lose confirmatory status.

The comparators. Three are confirmatory. Full replacement is the collapse condition and also the equal-budget random-selection control. The ten percent arm adapts the published retention control to this substrate, training each generation on 200 real rows freshly drawn from the 2,000 generation-0 real rows plus 1,800 candidates, fit by the same procedure as every other arm. A likelihood filter keeps the 2,000 most likely candidates under the current generator. Three are secondary. The equal-weight pool is the only contrast that can attribute a difference to the weights. A single verifier per tag measures pooling and weighting together. A bounded-exposure arm retains 192 fixed real rows every generation and is tested for non-inferiority only, because a keyed row and a real row need not carry the same information. Two reference arms gate interpretation: an oracle that accepts exactly the valid candidates, and a no-recursion arm trained on fresh real data, which defines whether collapse occurred.

The pool under test omits most of what makes RLRF an institution. It has no economy, no strategy, no adversary, no deliberation, and no reputation dynamics after the single settlement.

IV. Design: Three Substrates, One Frozen Packet

The design binds in one file, frozen by hash only after twenty-two rounds of blind audit by two model engines, the last of which both passed with no findings. Every harness and analysis step refuses to run

against a design whose hash differs. Every random draw derives from a named hash template, so no draw depends on anything outside its listed fields.

Substrate P, the replication gate. It replicates the published language-model protocol: the 125-million-parameter causal language model the published protocol specifies (Shumailov et al. 2024), fine-tuned on a standard Wikipedia text corpus (Merity et al. 2016), full replacement for five epochs against ten percent retention for ten, generations 0 through 9, five runs each. The filter never enters it. The frozen pass rule requires, on the five-run means from generation 0 to 9, a full-replacement perplexity rise $\Delta_{\mathrm{FR}} \geq 4.0$ with $\Delta_{\mathrm{FR}} - \Delta_{\mathrm{R10}} \geq 2.0$, where Δ_{R10} is the same rise under ten percent retention. The thresholds sit at fractions of the reported 8-point change so that the rule does not depend on which of two published baselines, 20 or 34, was meant. A miss blocks reading anything else.

Substrate G, the Gaussian mixture. Ten components in two dimensions, placed by hash at least 6 units apart on the square from -12 to 12. Four are assigned to the rare set by hash after placement, at weight 0.015 each. All four pre-run checks passed: validity balance, a rare-region mass of 0.060, rare cells of about 0.015 each, and tag leakage on the 160th center set. The study runs 200 seeds with a reserve of 40, fifty generations each. All 200 seeds completed in every family, with no fit failure and no reserve seed used. The outcomes sit in a sealed store.

Substrate F, declared procedure-invalid. A synthetic fact corpus was to be the fact-corpus test of the filter. The operator declared it procedure-invalid before any run or reading. No pre-run rule fired, and no written reason was recorded. Its nine registered tests stand at p equal to 1 with no other label, and the paper draws no inference from it. The consequence is plain: the filter is tested on one substrate, a two-dimensional mixture with simulated validators.

Two departures. The preregistered pilot, three seeds and two generations meant to confirm that every arm completes, was never run, and the confirmatory seeds ran without it. Their later completion does not cure the skipped gate. The substrate F declaration is the second departure. Both qualify every reading in Section VIII.

V. Estimands and Inference

The estimand. For the set $\mathcal{R}$ of four rare components, with $p_{50}(k)$ the probability the generation-50 model assigns to component k's cell and m_k that cell's frozen mass, the outcome is

$$R_{50} = \frac{1}{|\mathcal{R}|} \sum_{k \in \mathcal{R}} \min\left(1, \frac{p_{50}(k)}{m_k}\right).$$

Only generation 50 is confirmatory. Each final-generation estimate uses $L_G = 4 \times 10^6$ draws under common random numbers. With $m_{\mathrm{cell,min}} = 0.01$ the frozen floor that every rare-cell mass was required to clear before any run, that bounds each component's standard error by $0.5/(\sqrt{L_G}\, m_{\mathrm{cell,min}}) = 0.5/(2{,}000 \times 0.01) = 0.025$, a quarter of the smallest effect of interest. Uncapped rare mass is reported beside it, supports no claim, and shows what the cap changed.

Uncertainty and labels. The unit is the seed, and every contrast is a paired seed-level difference within a validator family. A seed bootstrap with $B = 10{,}000$ replicates gives percentile intervals and the two-sided p-value $p = \frac{1}{B}\,\#\{b : |d_b - d| \geq |d|\}$ for an observed contrast d, where d_b is the contrast in bootstrap replicate b (Efron and Tibshirani 1993). A contrast is superior when its Holm-adjusted p-value is below 0.05 (Holm 1979), its 95 percent interval lies above zero, and its point estimate satisfies $\hat{d} \geq \delta = 0.10$. It is negative when

the adjusted p-value is below 0.05 and the interval lies below zero, a substantive null when its 90 percent interval lies within $[-\delta, \delta]$, and inconclusive otherwise. The effect sizes, $\delta = 0.10$, a planning alternative $\delta_{\text{alt}} = 0.15$, and a non-inferiority margin $\delta_{\text{NI}} = 0.05$, were fixed by the operator before any pilot.

Families. The confirmatory family of twelve holds the pool against full replacement, the ten percent arm, and the likelihood filter in each of the three required families, plus the three substrate F tests at p equal to 1. Three secondary families hold the reputation-over-pooling, single-verifier, and non-inferiority contrasts. A failure-mode family of twelve holds the likelihood filter, the equal-weight pool, and the reputation pool, each against full replacement. Family Z sits outside every Holm family.

Gates. Before any label, any of the three confirmatory contrasts computed in family Z with a 95 percent interval above zero and $\hat{d} \geq 0.10$ is a leak, and a family Z minimum seed-mean fill-free share below 0.98 for the single verifier, the equal-weight pool, or the reputation pool is an acceptance failure. Either removes confirmatory status on both substrates. After labels, four conditions restrict claims. The collapse rung requires the no-recursion arm to beat full replacement by at least 0.10 with an interval above zero, and a planning calculation put that gap near 0.4, an expectation stated before any run, not a result. The oracle premise requires the oracle to clear the same bar in each family, and if it fails, no positive label is attributed to validity filtering. The coverage gate requires the pool's seed-mean coverage, the fraction of rare components retaining at least half their original mass, to be at least one half, with the 95 percent lower bound of its coverage minus full replacement's above zero. The key cap holds by construction at 192 against 200.

The ladder. A null is more useful when its location is known. Six rungs, read in order, report the first link not demonstrated among the five that carry a pass rule: collapse, the oracle premise, whether the weights tracked reliability on held-out keyed items, measured by a Spearman correlation (Spearman 1904) whose 95 percent lower bound must exceed 0.3, whether the weights improved the validity of what the pool accepted, a descriptive composition rung that never stops the ladder, and retention itself. A failed rung means a link was not demonstrated, not that the effect is absent.

Power. A positive reading on substrate G needs nine superior labels. For a contrast with paired seed-level standard deviation s_d over S seeds, $\text{se} = s_d/\sqrt{S}$. Under a normal approximation and a true effect δ_{alt}, the label power is

$$\Phi\left(\frac{\delta_{\text{alt}} - \max(\delta,\ z_{1-0.05/24}\,\text{se})}{\text{se}}\right), \qquad z_{1-0.05/24} \approx 2.865,$$

where $z_{1-0.05/24}$ is the critical value at the most demanding Holm step and Φ is the standard normal distribution function. The design takes the product of the nine single-test powers as the conjunction power. At $S = 200$ and $\delta_{\text{alt}} = 0.15$, the break-even s_d for a conjunction power of 0.80 is about 0.36, a figure that holds only when the comparator's retention is at most 0.85. If the ten percent arm's retention exceeds 0.90 in a family, the pool cannot be labeled superior to it there, and no positive reading is possible. No standard deviation was known before the runs, so the design fixes a masked release: after the gate opens, the analysis releases only each test's variance of paired differences, withholds the mean, and reports the realized power first. If the realized conjunction power is below 0.80, the substrate is reported as underpowered, with the same prominence as any label. That power covers the labels only, not the gates.

VI. Integrity Architecture

A preregistration binds an author only to the extent that departures from it would be visible. The audits came first. The design took twenty-two rounds. The substrate G harness and analysis were audited together

over five rounds, and the harness passed in the fifth with minor findings, which were queued for a later commit, to be audited before unsealing. The harness file has not changed since, and the confirmatory seeds ran under a later commit that differs from it only in documentation and the pre-run record. The analysis was then audited alone and passed with no findings in its thirteenth round, before any outcome was read. The substrate P harness passed in its seventh. Each of the ten sections of the longer working draft from which this text is condensed was audited by model engines before it was committed, with the exceptions Section IX notes. This condensed text is audited in its own rounds, and the record of those rounds is kept apart from those of the ten sections. The engines are not independent peer review: they may share failure modes with each other and with the author's tools.

Outcomes are written only to a sealed store, readable only by their owner, with a manifest of digests for every file. The seal is a commitment made visible, not a lock: the operator can open the files. The analysis checks each manifest's provenance, checks every sealed file against its digest before reading it, and records a digest of every file the variance release reads, refusing at unseal if any changed. It refuses unless the design hash is accepted, the tree is clean and matches its commit byte for byte, and the code differs from the audited commits only in a named allowlist. It scans for any module that could shadow its libraries before loading one, and it reads the audited commits it trusts from a local file outside version control, where no commit can move them. The store stays sealed until a substrate P record meets the frozen pass rule.

The design also fixes, before any data, the sentences the paper may write for each outcome and the conditions for each, and it bars claims about staking incentives, slashing, collusion, Sybil resistance, deployed systems, natural web text, frontier-scale models, or a solution to collapse. The architecture shows that the author arranged not to read the outcome early and that specified later adjustments would be detectable. It does not show that the filter works, and it does not prove the outcome was never read. The Gaussian outcomes also already exist, written before any venue reviewed the design, so the paper does not claim untouched Stage 1 status.

VII. Results

The replication verdict is reported first, because it governs everything after it. The record reports, for each arm, the five-run mean perplexity on the test split at generation 0 and at generation 9, and the rise between them. Full replacement: generation 0 42.22, generation 9 98.26, rise 56.04. Ten percent retention: generation 0 43.15, generation 9 330.44, rise 287.28. Full replacement's rise minus the ten percent arm's rise: -231.24.

The pass rule requires a full-replacement rise of at least 4.0 and a rise at least 2.0 larger than the ten percent arm's. If the rule is not met: the published collapse result did not reproduce under the frozen rule. The replication failed. Substrate G is not read, its outcomes stay sealed, substrate F is blocked as well, and the paper reports no filter result. Rereading the same output under a different statistic is barred, and any further attempt requires a new version of the design and a new blind audit.

The preregistered pilot was not run, and the confirmatory seeds on substrate G ran without it. Their later completion does not cure the skipped gate. Substrate F is reported as procedure-invalid. It was declared so by the operator before any run or reading, and no written reason was recorded when the declaration was made. No pre-run rule of the frozen design fired. Its nine registered tests stand at $p = 1$ with the procedure-invalid label and no other. In the confirmatory family they are the pool against full replacement, against ten percent retention, and against the likelihood filter. In the three secondary families they are the reputation pool against the equal-weight pool, the reputation pool against the single verifier, and the non-inferiority test against the bounded-exposure arm. In the failure-mode family they are the likelihood filter, the equal-weight pool, and the reputation pool, each against full replacement. No inference is drawn from it.

The paper does not claim anything about staking incentives, slashing, collusion, Sybil resistance, deployed systems, natural web text, or frontier-scale models, and it does not claim that collapse is solved.

VIII. What Each Outcome Would License

If substrate P does not meet its frozen pass rule, the paper reports that the published collapse result did not reproduce under the frozen rule, that substrate G is not read, that its outcomes stay sealed, that substrate F is blocked as well, and that no filter result is reported. The reading concerns the study's implementation of the published language-model protocol, including the choices the design fixed where the published methods left them open. The study would not be in a position to say whether the implementation, those choices, or the published result accounts for the miss. The consequence is procedural: rereading the same output under a different statistic is barred, and any further attempt requires a new version of the design and a new blind audit. On this outcome the study examines nothing about the filter.

This reading carries the two disclosed departures. The preregistered pilot was not run, and substrate F was declared procedure-invalid by the operator before any run or reading. No pre-run rule of the frozen design fired, and no written reason was recorded when the declaration was made. Its nine registered tests stand at $p = 1$ with the procedure-invalid label and no other.

IX. Limits

The filter is tested on one substrate, a two-dimensional mixture, and never inside a language-model loop. Its validators are simulated, and the families are, in the design's words, the result. They err independently given their accuracy, which depends only on the tag, so shared lineage, correlated errors, and error that depends on whether content is rare are excluded by construction, and the study supplies no evidence about correlated assessors or about reputation concentrating in a lineage. Validity is balanced between rare and common content by construction, so the study says nothing about settings where rare content is harder to check. The weight is settled once and compared only with equal weighting and a single verifier, never with a key-free aggregator (Dawid and Skene 1979). Stake has no external value, and the study cannot speak to incentives, collusion, Sybil resistance, or Goodhart pressure. Nine superior labels set a high bar, so a real but modest effect may be reported as not positive. The two departures, the skipped pilot and the operator's declaration on substrate F, qualify every reading. And in the final audit rounds of the last three sections of the working draft, one audit family's command-line engines returned no verdict because their usage balance was exhausted, so those rounds hold two passing verdicts rather than three, and they left identified low-severity findings unadopted, as their audit records state.

Post-result disclosure: After the result, a read-only review of the ten percent arm found no implementation error and confirmed that the arm matched the frozen design. Two auditor engines confirmed the review and added findings. The published protocol itself specifies twice as many epochs for the ten percent setting, and one confirming audit notes that the published study reported only minor degradation under that split. One confirming audit finds that three choices can reverse the comparison between full replacement and the ten percent arm: retained real data was added to a full synthetic set, while the public code keeps the original size and replaces the other rows with continuations, later generations were scored at their final epoch while generation 0 was scored at its best validation epoch, and the design fixed the learning rate schedule where the published methods left it open. Neither the review nor the audits established that these choices caused the miss. The review licenses no claim about the filter and does not alter the Section VII reading.

The design assigns stake size, slashing cost, and collusion to a second study. The paper would add tests of

strategic adaptation and Goodhart pressure, a test inside a language-model loop on natural text, a comparison with a key-free aggregator and with filters that select on surprise or entropy, and later work on scale. None is promised. Each would need its own design, preregistration, and audit.

One follow-on study would bear most directly on the parent paper, by combining three changes, each aimed at a limit stated above. It would replace the simulated validators with model validators drawn from several lineages and settled on keyed items, so that correlated errors and the concentration of reputation in a lineage could be measured rather than excluded by construction. It would add, beside the keyed pool, an arm that reproduces the relevant elements of the parent paper's proposed answer-generation format. In that arm, reputation would update against realized pool resolutions on unkeyed events, and candidates that received an UP resolution with high reputation-weighted consensus would become training targets. Held-out keyed items would then measure whether the resulting weights track accuracy or only agreement with the pool, which is the parent paper's second open problem, under recursion. And it would run the filter inside a language-model loop on natural text. Such a study would still omit economic stake and deliberation, and so would test the parent mechanism only closer to its published form, not as a whole. Nor would it be the end-to-end policy comparison the parent paper names as prior to all six of its numbered open problems, which would remain a separate study. This follow-on is the paper's suggestion and is separate from the second study that the design names, which would take up stake size, slashing cost, and collusion. Like the items above, it is not promised.

X. Conclusion

The question was never whether reputation can be described as a fitness function for recursive training. It is whether a domain-scoped reputation weight, settled once against a key, measurably improves a validity filter under recursive replacement, on the substrate where the test can be run. The paper contributes a way of asking: a ceiling named before any data, an outcome that counts the death of a rare component, a separation of what a pool does from what its weights do, sentences fixed before the outcome, and a record of where the study departed from its own procedure.

The published collapse result did not reproduce under the frozen rule. The replication failed. Substrate G is not read, its outcomes stay sealed, substrate F is blocked as well, and the paper reports no filter result. The study would not be in a position to say whether the implementation, the choices the design fixed where the published methods left them open, or the published result accounts for the miss. Rereading the same output under a different statistic is barred, and any further attempt requires a new version of the design and a new blind audit. On this outcome the study examines nothing about the filter, and the narrow hypothesis remains unexamined.

A mechanism that has not been tested has not been shown to work.

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